Plan a prediction in two dimensions

LEARNING TARGET I can plan a prediction using evidence and fixed axes.

SECTION A KEY VOCABULARY

Frame	Keep the origin and axes fixed.
Components	Signed x and y parts; calculate separately.
Units	Make all units match before substituting.
Limit	State assumptions, such as steady velocity.

SECTION B KEY EQUATIONS + UNITS

$$x_f = x_0 + v_x t$$

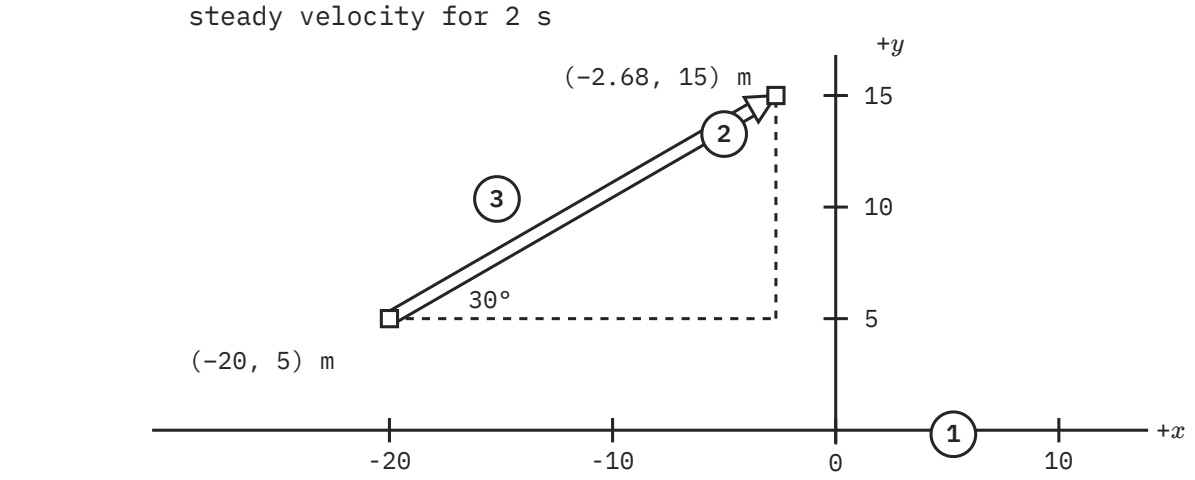
$$y_f = y_0 + v_y t$$

v_x $v \cos\theta$

v_y $v \sin\theta$; θ from $+x$

SECTION C COLOR IT · START + CHANGE

Probe example; fixed axes.



☐ **1 · PENCIL** Origin and axes

☐ **2 · BLUE** Start and end positions

☐ **3 · BLUE** Displacement direction

SECTION D WORKED EXAMPLE

A probe starts at (−20, 5) m and moves at **10 m/s**, **30°** from $+x$ toward $+y$, for **2 s**. **Find its new coordinates.**

GIVEN	EQUATION	WORK	ANSWER
x_0 = **−20 m**; y_0 = **5 m** v = **10 m/s**; θ = **30°** t = **2 s**; steady v	Resolve v: v_x = **8.66 m/s** v_y = **5 m/s**	x_f = −20 + 8.66(2) y_f = 5 + 5(2)	**(−2.68, 15) m** Final x is still negative.

SECTION E THE 2026-XJ THREAD

DAY 21 OF 22 · NEW TOOL: A TWO-AXIS PLAN

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else